

Machine Learning Assessment (Version 1)

The student must apply machine learning techniques to solve a real-world problem using this data. [Customer Segmentation Classification](#)

The main objectives are to:

- Understand the full machine learning workflow.
- Preprocess and explore data.
- Build and compare machine learning models.
- Evaluate results using suitable metrics.
- Explain findings and limitations

The task should include the following:

- Suitable data preprocessing (e.g., handling missing values, remove duplications,...)
- Exploratory data analysis
- Feature engineering pipeline
- Model training (i.e., at least 4 models)
- Model evaluation (using various metrics including the confusion matrix)
- Comparison of models' results and ROC curves
- Optional (hyperparameter tuning)
- Result discussion

Please, submit a clear and organized notebook.